MASTER OF SCIENCE
IN ENGINEERING

Spectrum estimation

Statistical Signal Processing and Modelling

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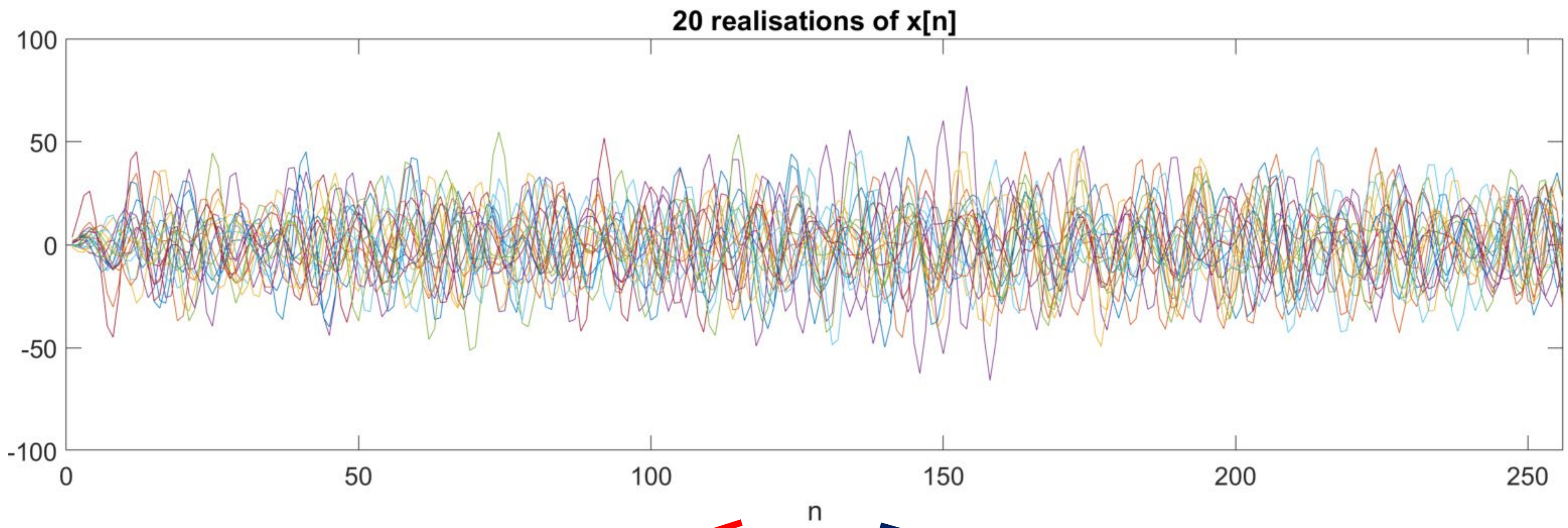
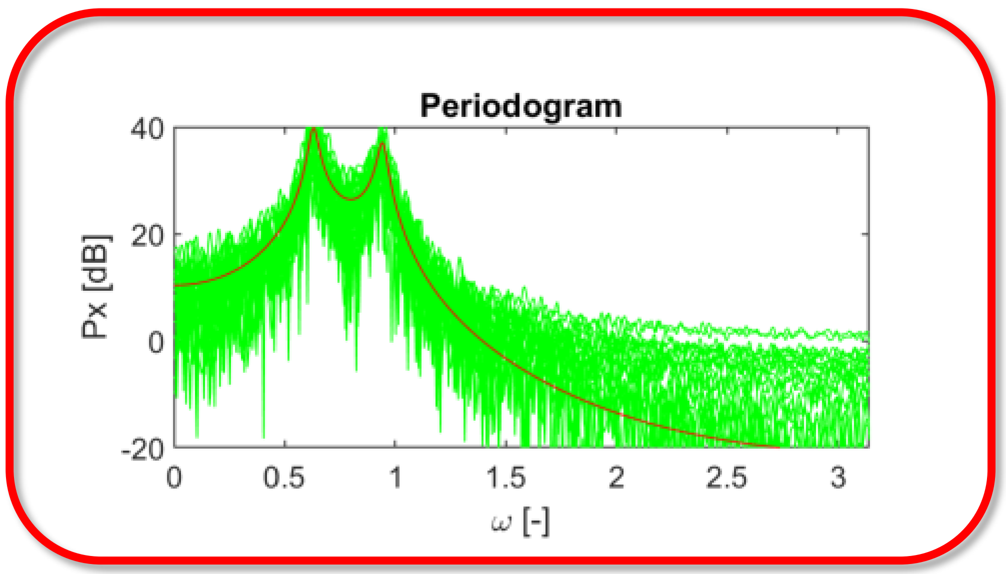
Introduction

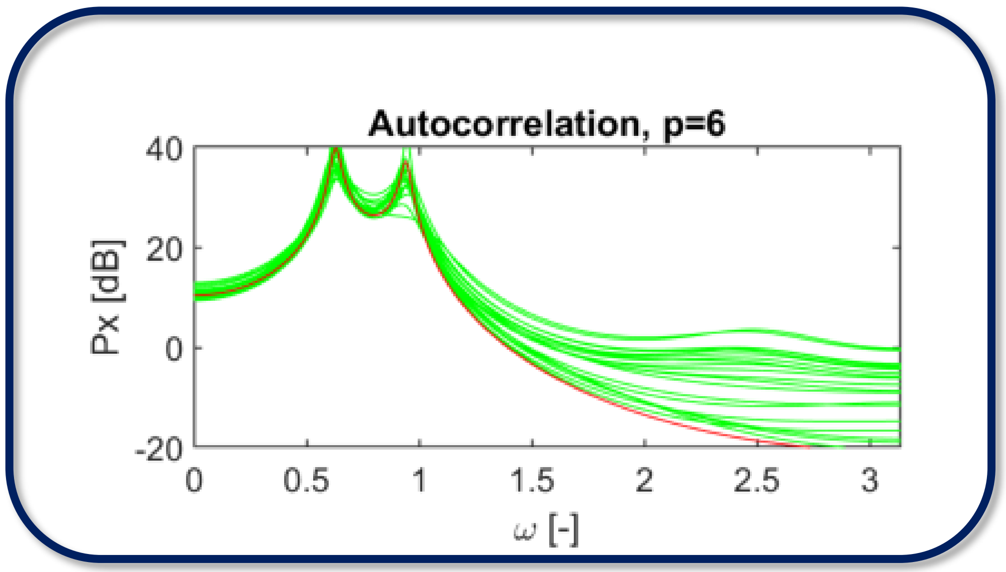
- There are 2 main ways to estimate the spectrum of a process



Non-Parametric

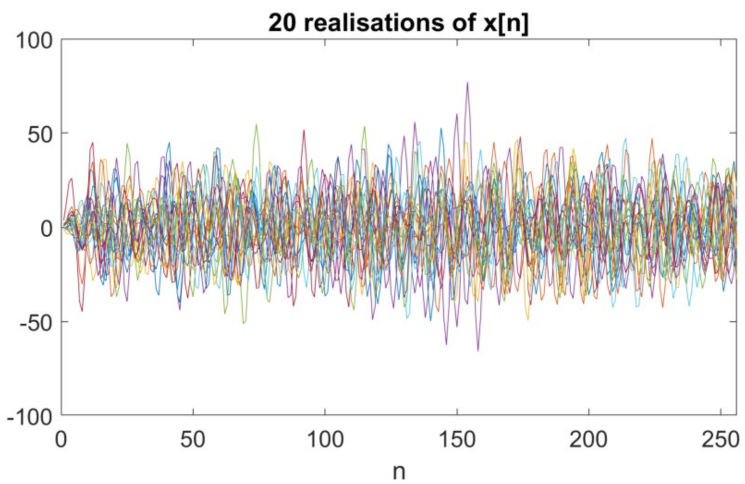


Parametric



Introduction

- The parametric method is using the signal modeling technics:



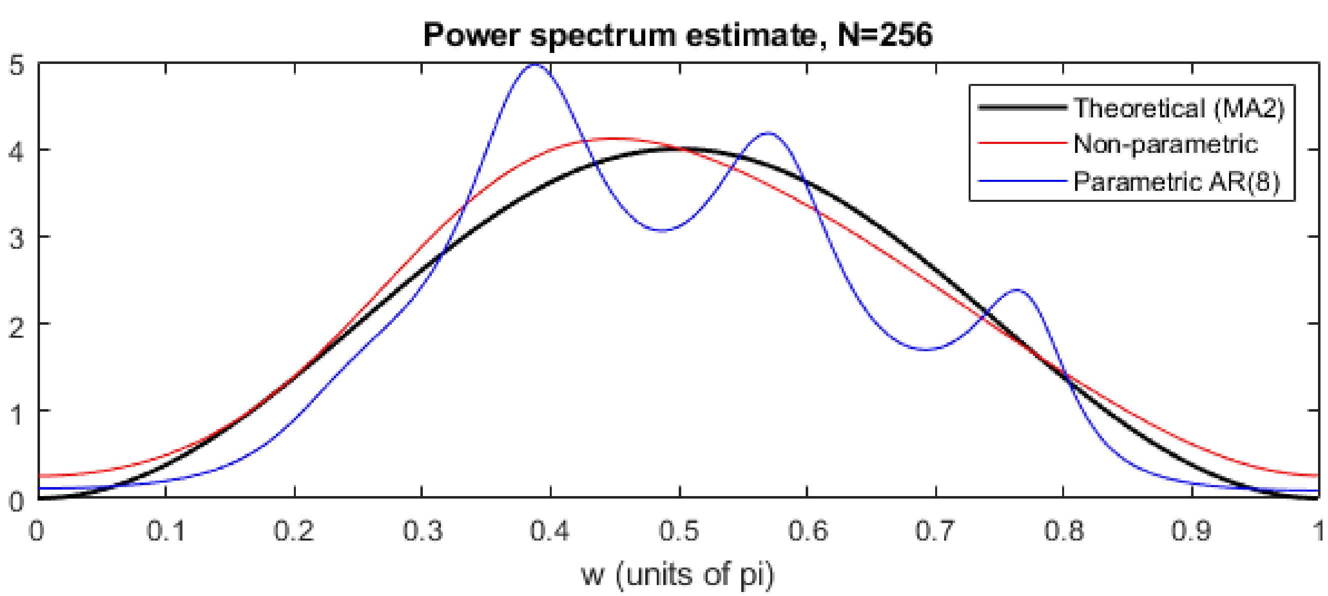
$x[n]$ → Signal modeling (AR / MA / ARMA) →

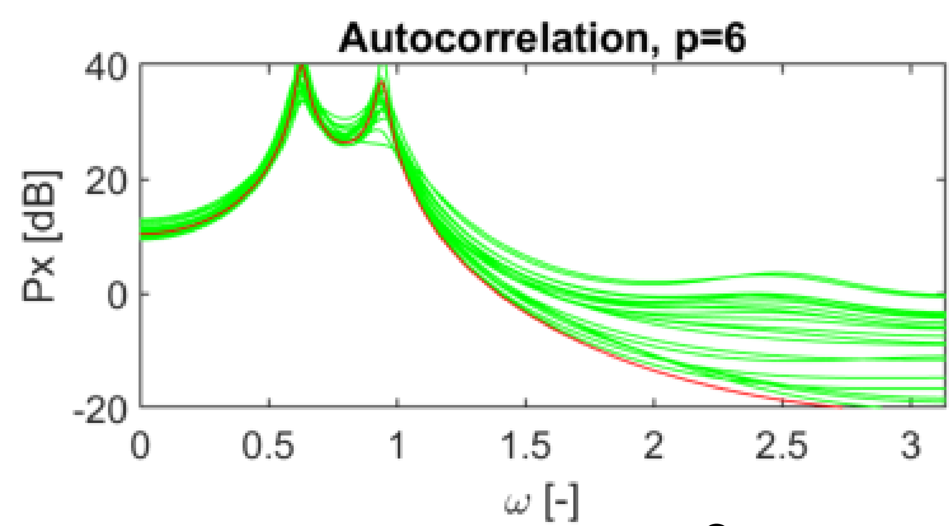
$$X(z) = \frac{\sum_{k=0}^{q} b[k] z^{-k}}{1 + \sum_{k=1}^{p} a[k] z^{-k}}$$

$z = e^{j\omega}$

$$\mathrm{P}_x(e^{j\omega}) = \frac{\left|\sum_{k=0}^{q} b[k] e^{-jk\omega}\right|^2}{\left|1 + \sum_{k=1}^{p} a[k] e^{-jk\omega}\right|^2}$$

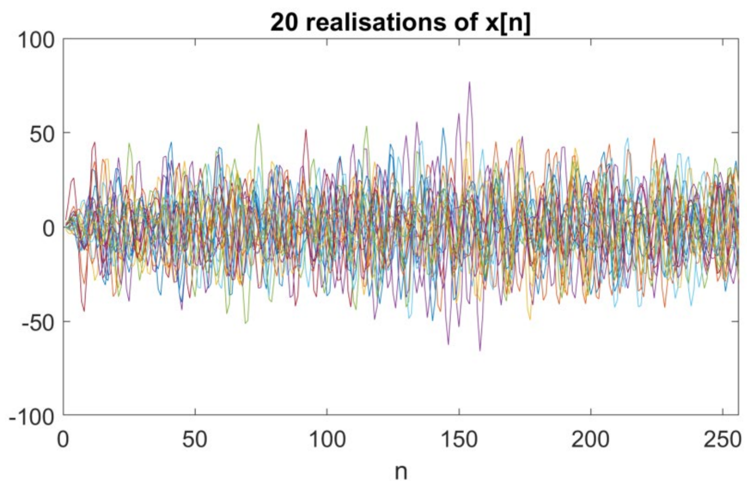
- Can integrate process information
- Better to reject noise if process can be estimated
- Various method to modelize (AR/MA/ARMA)
- Not generic approach
  - For example, AR model is not good to estimate MA process



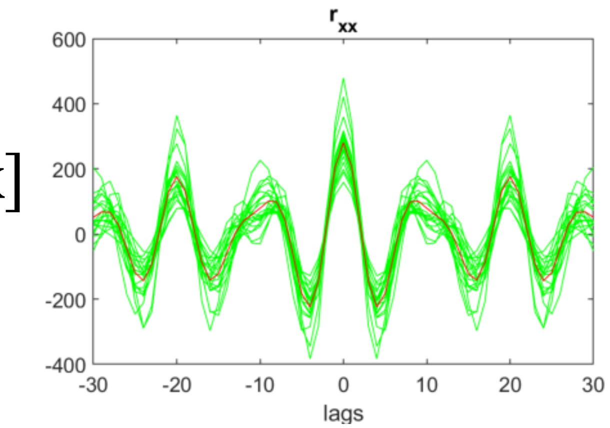


Introduction

- <u>The non-parametric method</u> is using Fourier and/or autocorrelation:

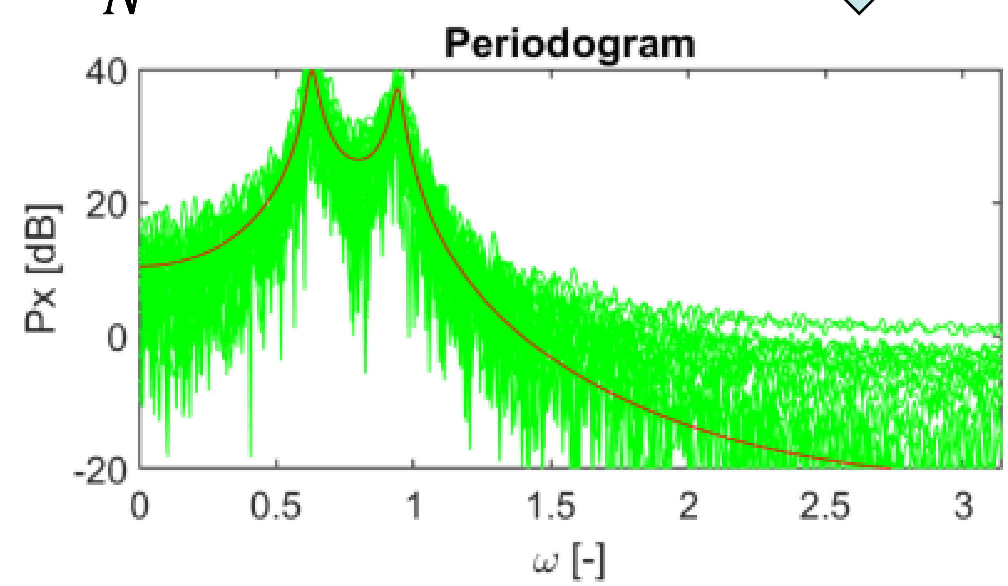


$x[n]$ → Auto-correlation → $r_X[\mathrm{k}]$



$x[n]$ → Fourier → $P_x = \dfrac{|X(f)|^2}{N}$

$r_X[\mathrm{k}]$ → Fourier → P_x



- Generic approach
- Estimating the power spectrum is equivalent to estimate the autocorrelation
- Easy to compute but limited to produce accurate power spectrum, especially for short data records.
- Various methods :
  - periodogram, modified periodogram, Barlett's, Welch's, Blackman-Tukey

Introduction

- The **Power Spectrum Density (PSD)** is in $\left[\frac{V^2}{Hz}\right]$ of a signal in $[V]$.
- It has to be integrated between 2 frequencies f_1 and f_2 with $f_1 < f_2$ to get an estimate of the signal's power between these 2 frequencies.
- The power spectrum density is defined by:

$$P_x(\omega) = \sum_{k=-\infty}^{\infty} r_{xx}[k]\, e^{-j\omega k}$$

Where $r_{xx}[k]$ is the autocorrelation function defined as:

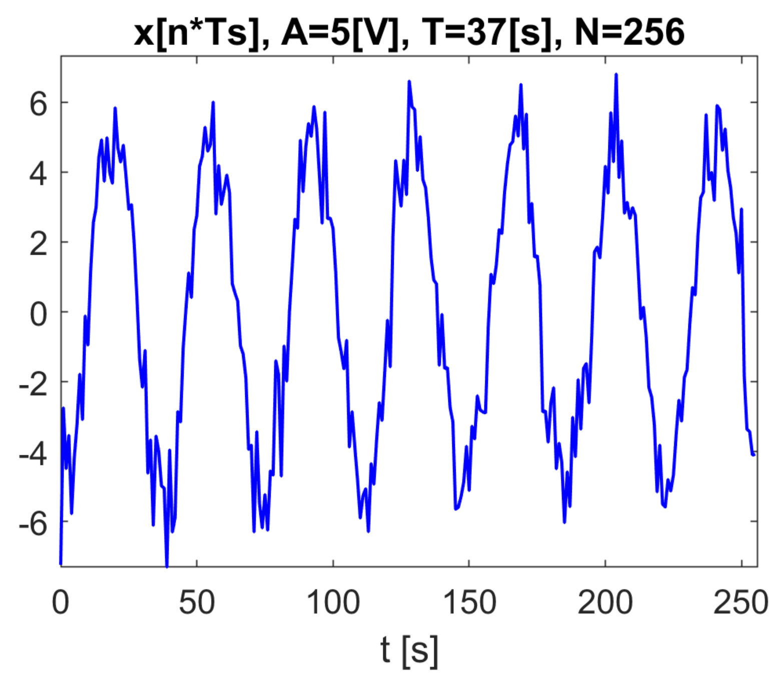
$$r_{xx}[k] = E\{x[l]x[l+k]\}$$

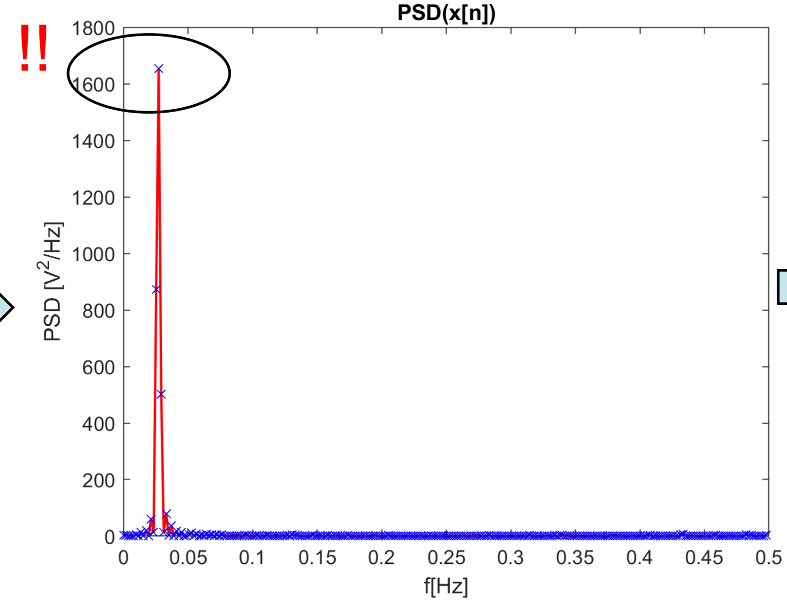
- Problem of estimating PSD is equivalent to estimate autocorrelation. Estimation could be biased when the data record is short.

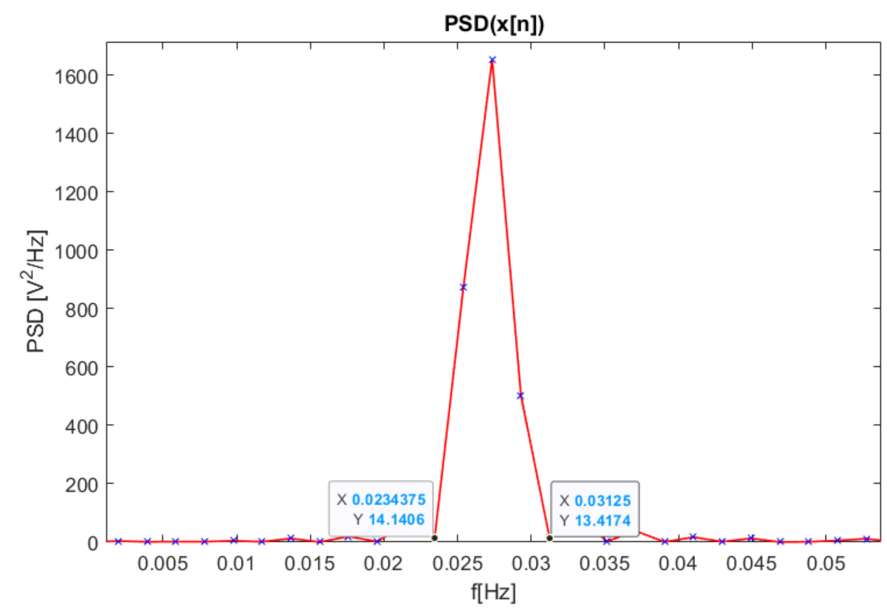
Introduction

- It is well known that a signal with amplitude $A[V]$ has a **power** of $\frac{A^2}{4}\ [V^2]$ in bilateral spectrum. Let's see.
  - Noisy sinus A= 5V → Power = $\frac{25}{4} = 6.25\ V^2$
  - T=37s, F0=1/T=0.027Hz, Fs = 1 Hz, Nfft = 512, df = 0.002 Hz

Signal **PSD**





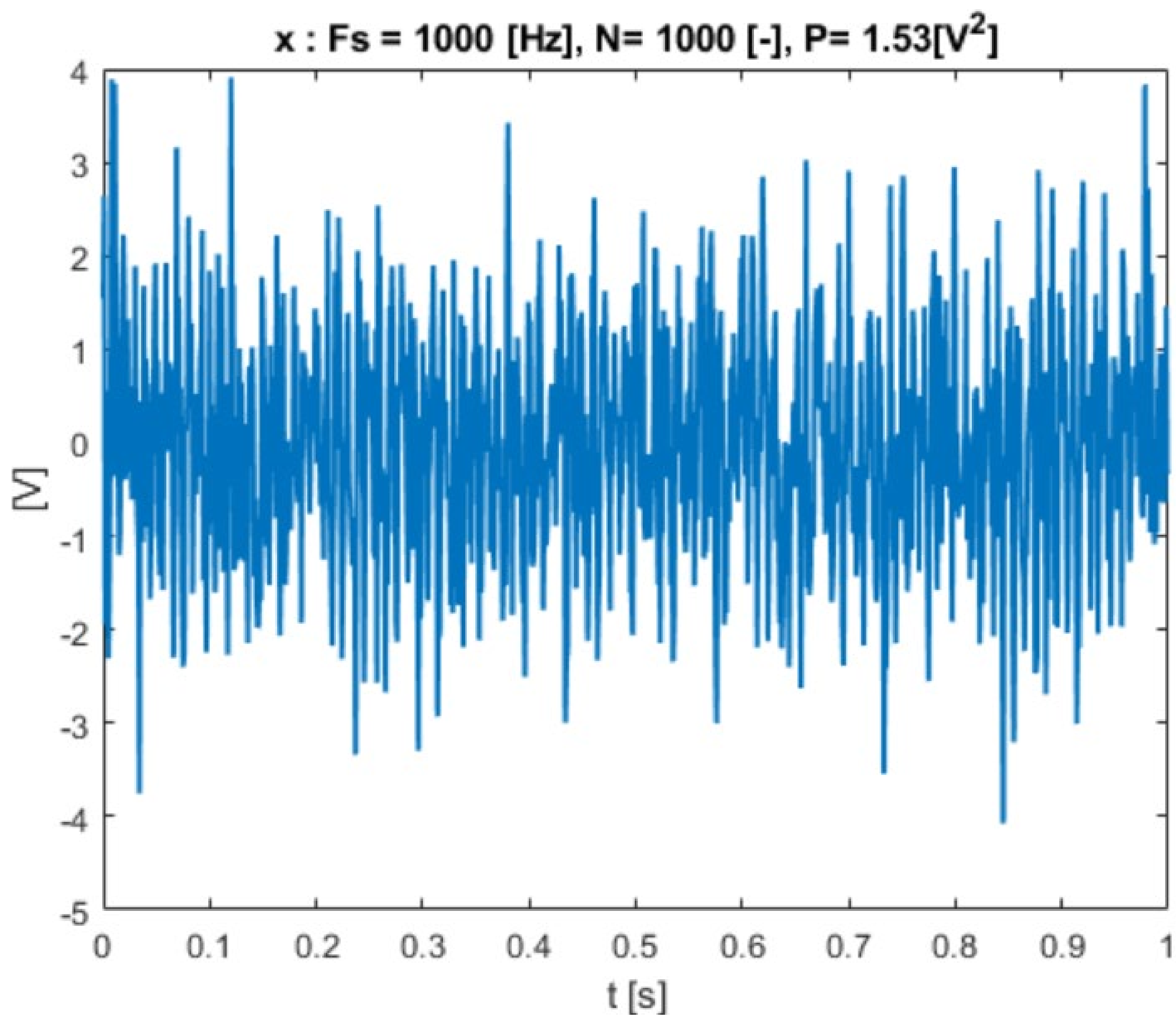


$$x[n] \quad \Rightarrow \quad \hat{P}_x = \frac{1}{N}\left|\sum_{k=0}^{N-1} x[k]\, e^{-j\omega k}\right|^2 \quad \Rightarrow \quad \sum_{f=0.02}^{0.03} \hat{P}_x(f)\cdot df \approx 6.1$$

Normalization

- Be careful, the psd is changing in function of the frequency domain representation.

This signal is a cosinus of 100Hz with a normalized gaussian noise



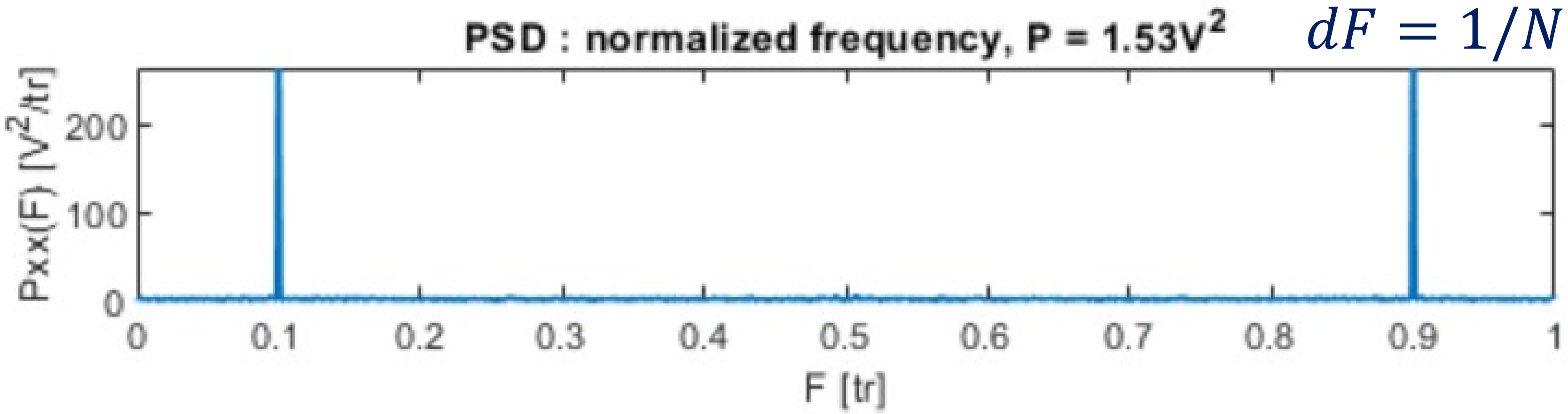
Normalization

- There are three frequencies domain to show the power spectrum density. But the total power is the same.

$$P_{xx}(F) = \sum_{k=-\infty}^{\infty} r_{xx}[k]\, e^{-j\omega k}$$

$$P = \int_0^1 Pxx(F) \cdot dF$$

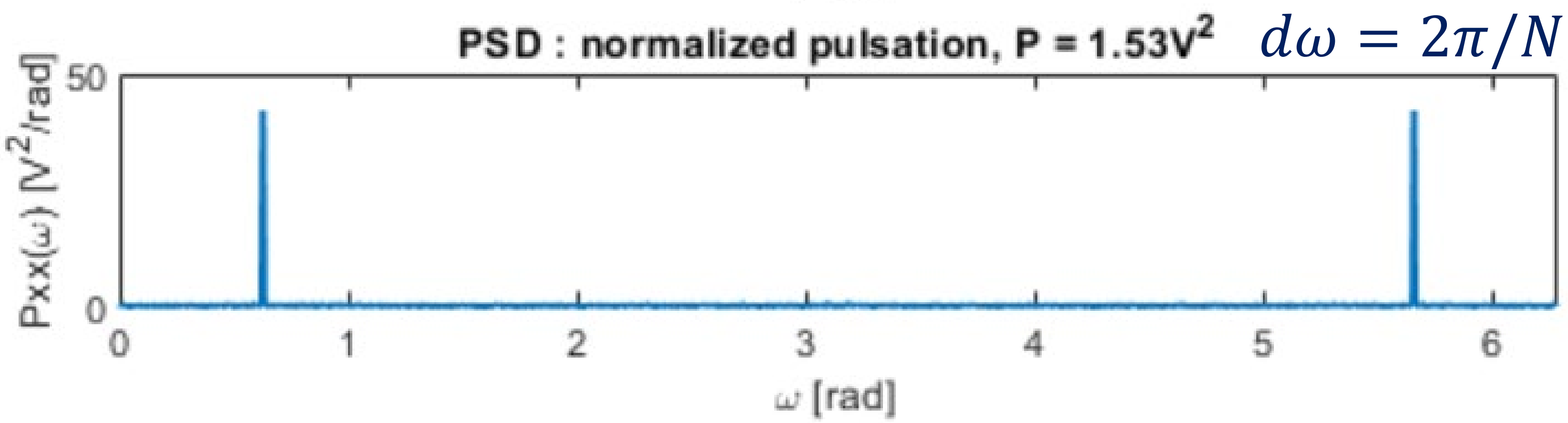
$$dF = 1/N$$



$$P_{xx}(\omega) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} r_{xx}[k]\, e^{-j\omega k}$$

$$P = \int_0^{2\pi} Pxx(\omega) \cdot d\omega$$

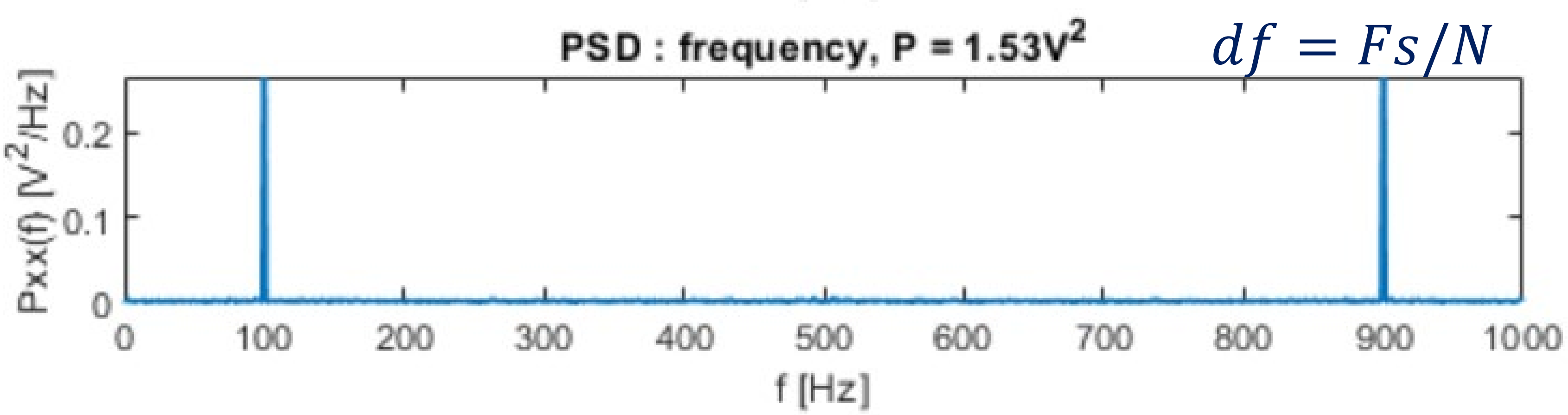
$$d\omega = 2\pi/N$$



$$P_{xx}(f) = \frac{1}{Fs} \sum_{k=-\infty}^{\infty} r_{xx}[k]\, e^{-j\omega k}$$

$$P = \int_0^{Fs} Pxx(\omega) \cdot df$$

$$df = Fs/N$$

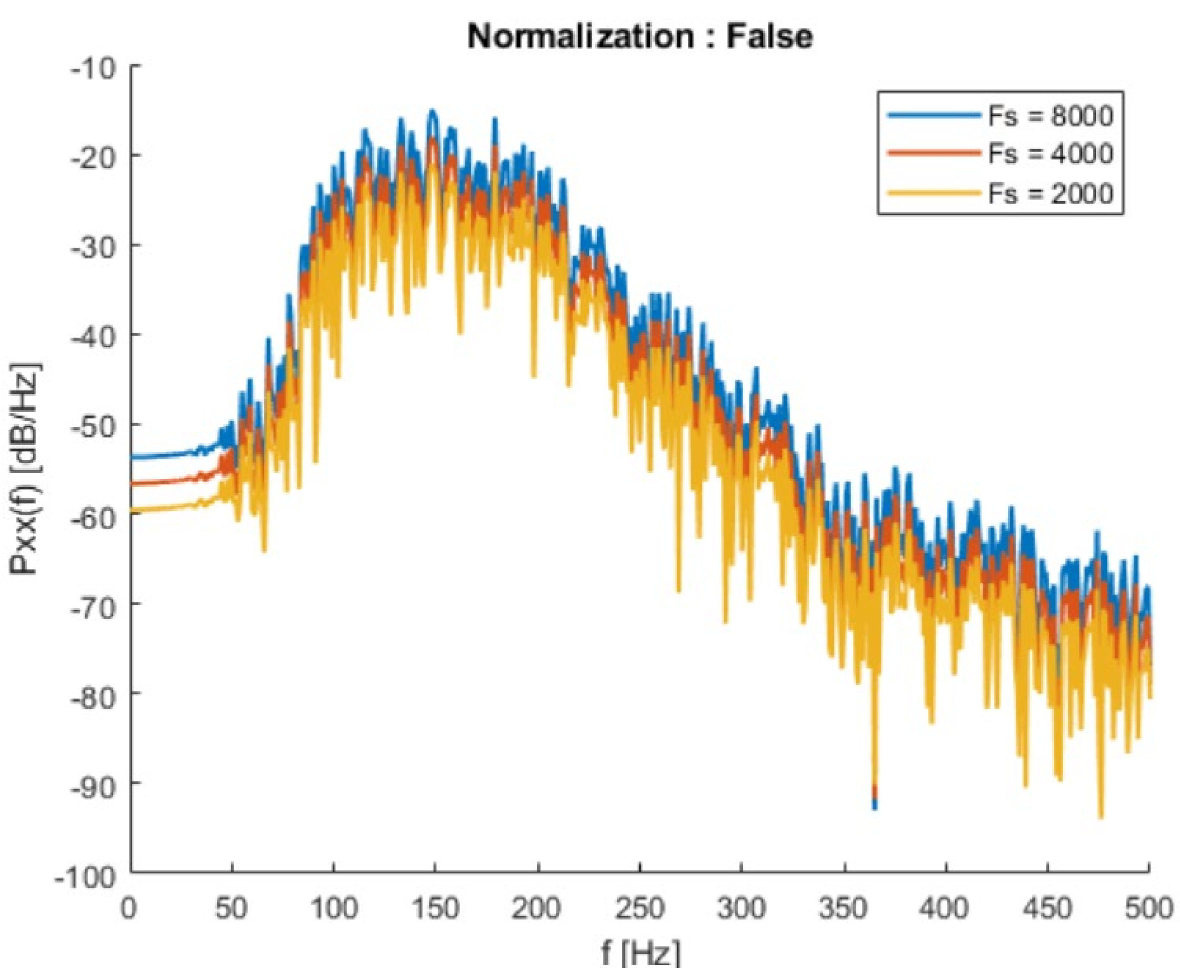


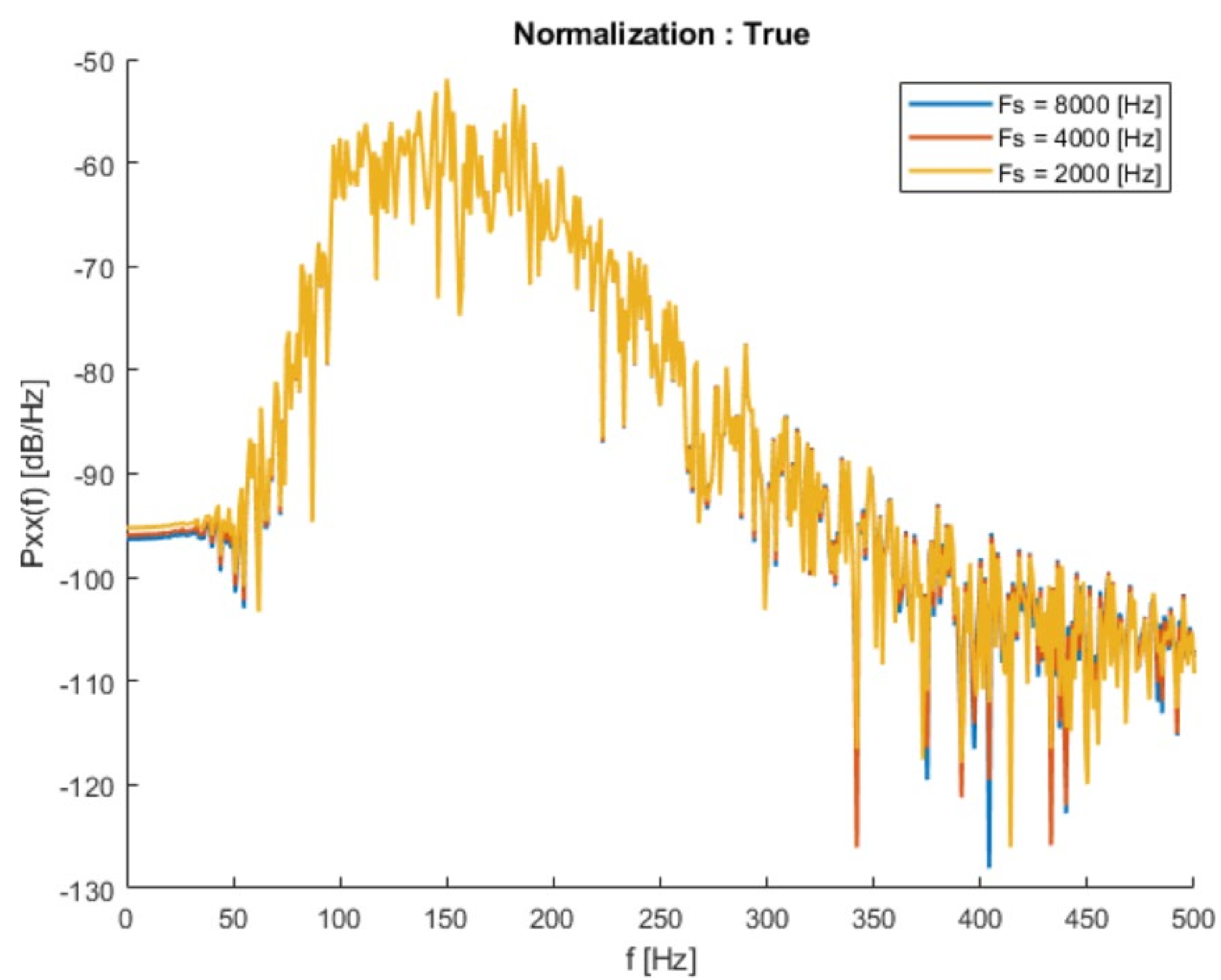
Normalization

- The frequency sampling can as well change the power spectrum density without normalization

$$P_{xx}(f) = \sum_{k=-\infty}^{\infty} r_{xx}[k]\, e^{-j\omega k}$$

$$P_{xx}(f) = \frac{1}{Fs} \sum_{k=-\infty}^{\infty} r_{xx}[k]\, e^{-j\omega k}$$





Periodogram

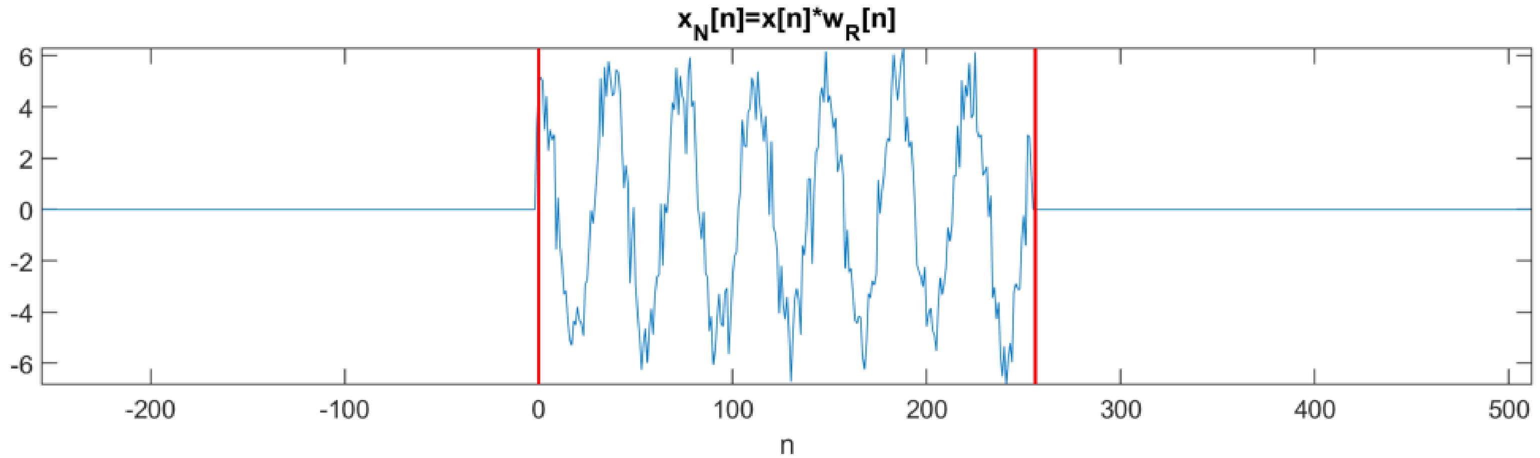
- **Periodogram** is the simplest estimator of the PSD:

$$P_x(e^{j\omega}) = \sum_{k=-\infty}^{\infty} r_{xx}[k]\, e^{-j\omega k} \quad \Rightarrow \quad \hat{P}_{per}(e^{j\omega}) = \sum_{k=-N+1}^{N-1} \hat{r}_{xx}[k]\, e^{-j\omega k}$$

- Therefore, spectrum estimation is, in some sense, an autocorrelation estimation problem.

Periodogram

- As the signal $x[n]$ is limited, it can be seen as $x_N[n]$ as the product of $x[n]$ with a rectangular window $w_R[n]$



$$x_N[n] = \begin{cases} x[n] & 0 \leq n < N \\ 0 & otherwise \end{cases}$$

$$\Rightarrow \quad x_N[n] = w_R[n] \cdot x[n]$$

- The estimated autocorrelation become :

$$\hat{r}_{xx}[k] = \frac{1}{N} \sum_{n=0}^{N-1} x[n+k]x^*[n]$$

$$\hat{r}_{xx}[k] = \frac{1}{N} \sum_{n=-\infty}^{\infty} x_N[n+k]x_N^*[n] = \frac{1}{N} x_N[k] * x_N^*[-k]$$

Periodogram

- Taking the Fourier transform and using the convolution theorem $F(f * g) = F(f) \cdot F(g)$, the periodogram becomes:

$$\hat{P}_{per}\left(e^{j\omega}\right) = \sum_{k=-N+1}^{N-1} \hat{r}_{xx}[k]\, e^{-j\omega k} = \frac{1}{N} \sum_{k=-\infty}^{\infty} \left(x_N[k] * x_N^*[-k]\right) e^{-j\omega k}$$

$$= \frac{1}{N} X_N\left(e^{j\omega}\right) X_N^*\left(e^{j\omega}\right)$$

Where:

$$X_N\left(e^{j\omega}\right) = \sum_{n=-\infty}^{\infty} x_N[n] e^{-jn\omega} = \sum_{n=0}^{N-1} x[n] e^{-jn\omega}$$

- Thus, the periodogram is simply the square magnitude of the DTFT of $x_N[\mathrm{n}]$

$$\Rightarrow \quad \boxed{\hat{P}_{per}\left(e^{j\omega}\right) = \frac{1}{N} \left|X_N\left(e^{j\omega}\right)\right|^2}$$

Periodogram example

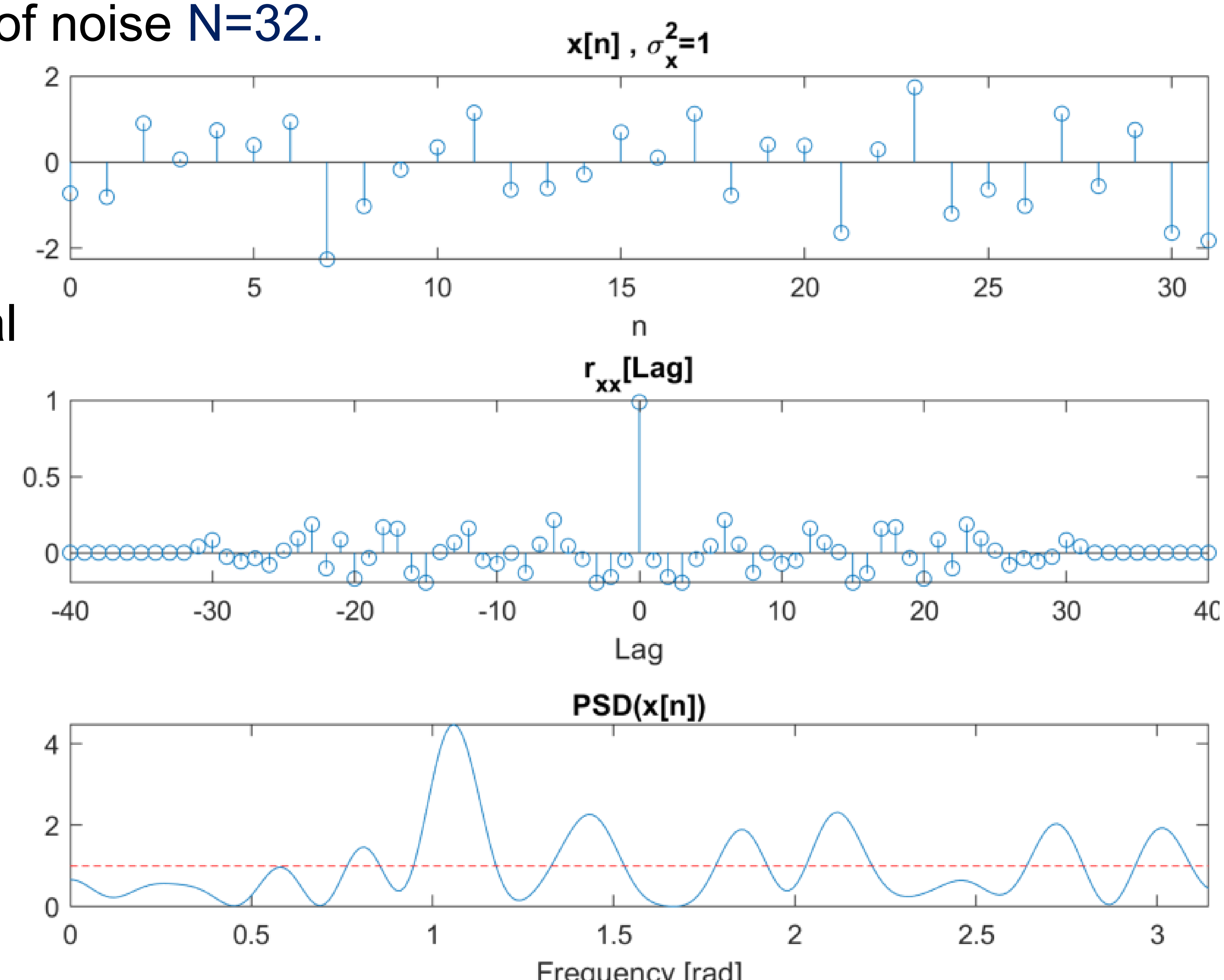
- If $x[n]$ is white noise with a variance of σ_x^2, then $\mathrm{r_x}[\mathrm{k}] = \sigma_x^2 \delta[k]$ and then the power spectrum is a constant

$$P_x\left(e^{j\omega}\right) = \sigma_x^2$$

- Next is showing one realization of noise N=32.

  - Estimated $\hat{r}_x[\mathrm{k}]$ is not a $\sigma_x^2 \delta[k]$
  - $\hat{P}_{per}\left(e^{j\omega}\right)$ is approximately equal to $P_x\left(e^{j\omega}\right)$ on the average
  - Considerable variation in $\hat{P}_{per}\left(e^{j\omega}\right)$ as ω varies.

⇨ Let's evaluate the performance of the periodogram!



Periodogram (performance)

- The unique parameter of the periodogram is N, the observation time.
- Ideally, as N increase, the periodogram should converge to the power spectrum **(bias):**

$$\lim_{N\to\infty} E\{\hat{P}_{per}(e^{j\omega})\} = P_x(e^{j\omega})$$

- And the **variance** that goes to zero as the data record length goes to infinity

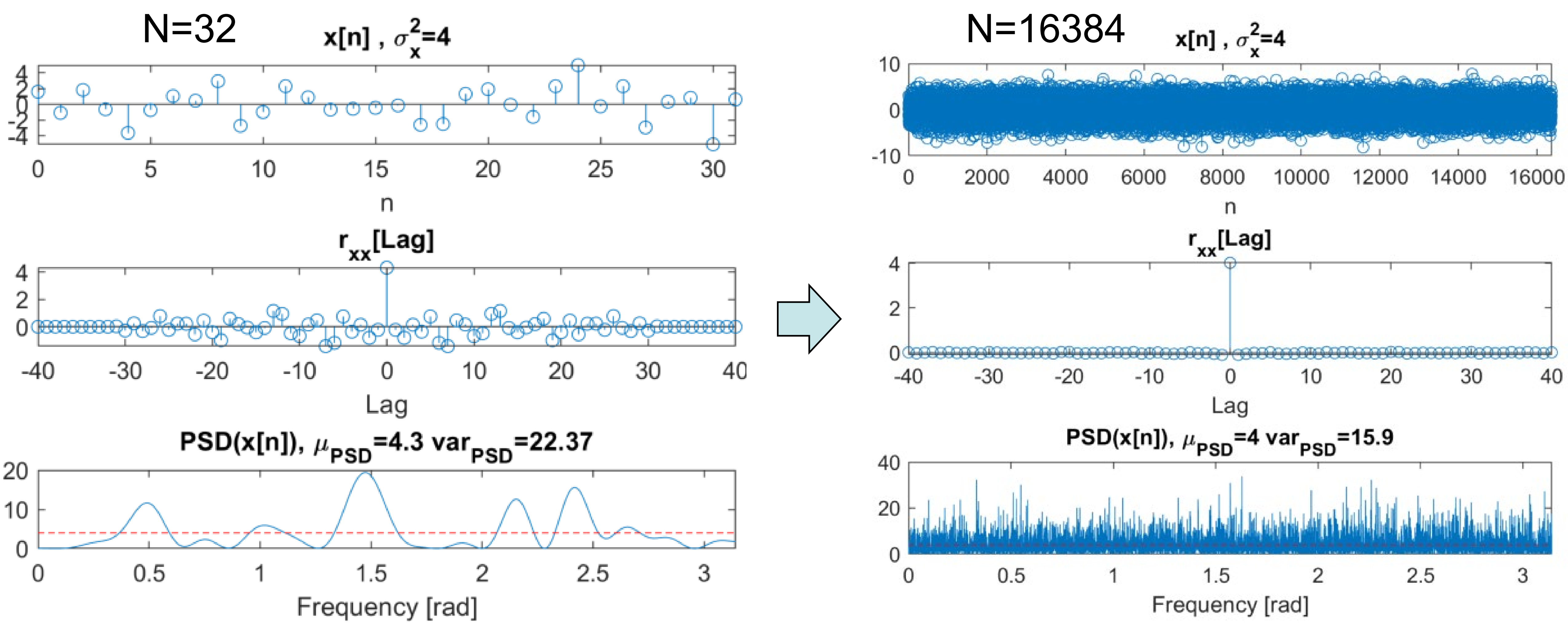
$$\lim_{N\to\infty} Var\{\hat{P}_{per}(e^{j\omega})\} = 0$$

- In other words, $\hat{P}_{per}(e^{j\omega})$ must be **a consistent estimate** of the power spectrum.

⇨ Let's evaluate the Bias and Variance of the periodogram!

Periodogram (performance)

- $x[n]$ is white noise with a variance of $\sigma_x^2 = 4$
- Increasing N is making the average correct $\left(E\left\{\hat{P}_{per}\left(e^{j\omega}\right)\right\} \approx P_x(e^{j\omega})\right)$ but the variance is still high $Var\left\{\hat{P}_{per}\left(e^{j\omega}\right)\right\} \approx P_x^2(e^{j\omega})$!
- The variance is not as good as expected.



Periodogram (Bias)

- Let's evaluate the periodogram bias mathematically:

$$E\{\hat{P}_{per}(e^{j\omega})\} = E\left\{\sum_{k=-N+1}^{N-1} \hat{r}_{xx}[k]\, e^{-j\omega k}\right\} = \sum_{k=-N+1}^{N-1} E\{\hat{r}_{xx}[k]\}e^{-j\omega k}$$

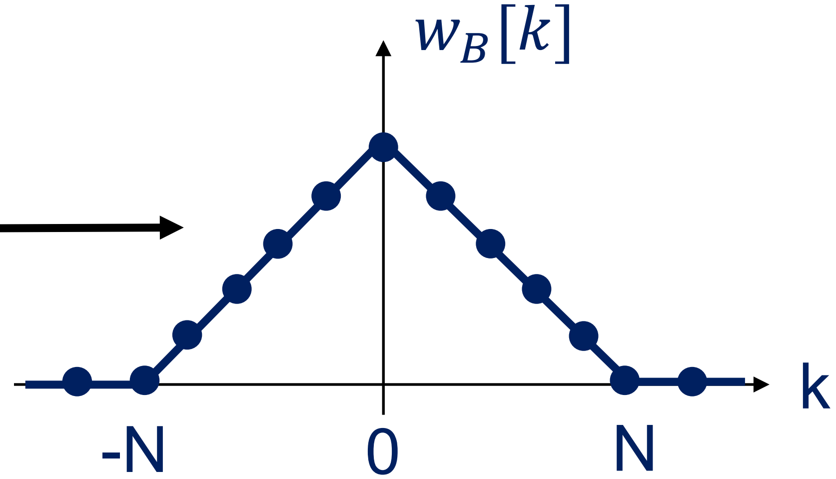
$$E\{\hat{r}_{xx}[k]\} = \frac{1}{N}\sum_{n=0}^{N-1-k} E\{x[n+k]x^*[n]\} = \frac{1}{N}\sum_{n=0}^{N-1-k} r_x[k] = \frac{N-k}{N} r_x[k]$$

$$k = 0,1,\dots,N-1$$

- So, it can be written: $E\{\hat{r}_{xx}[k]\} = w_B[k] \cdot r_x[k]$

$$w_B = \begin{cases} \dfrac{N-|k|}{N} & |k| \le N \\ 0 & |k| > N \end{cases}$$

Barlett Window



Periodogram (Bias)

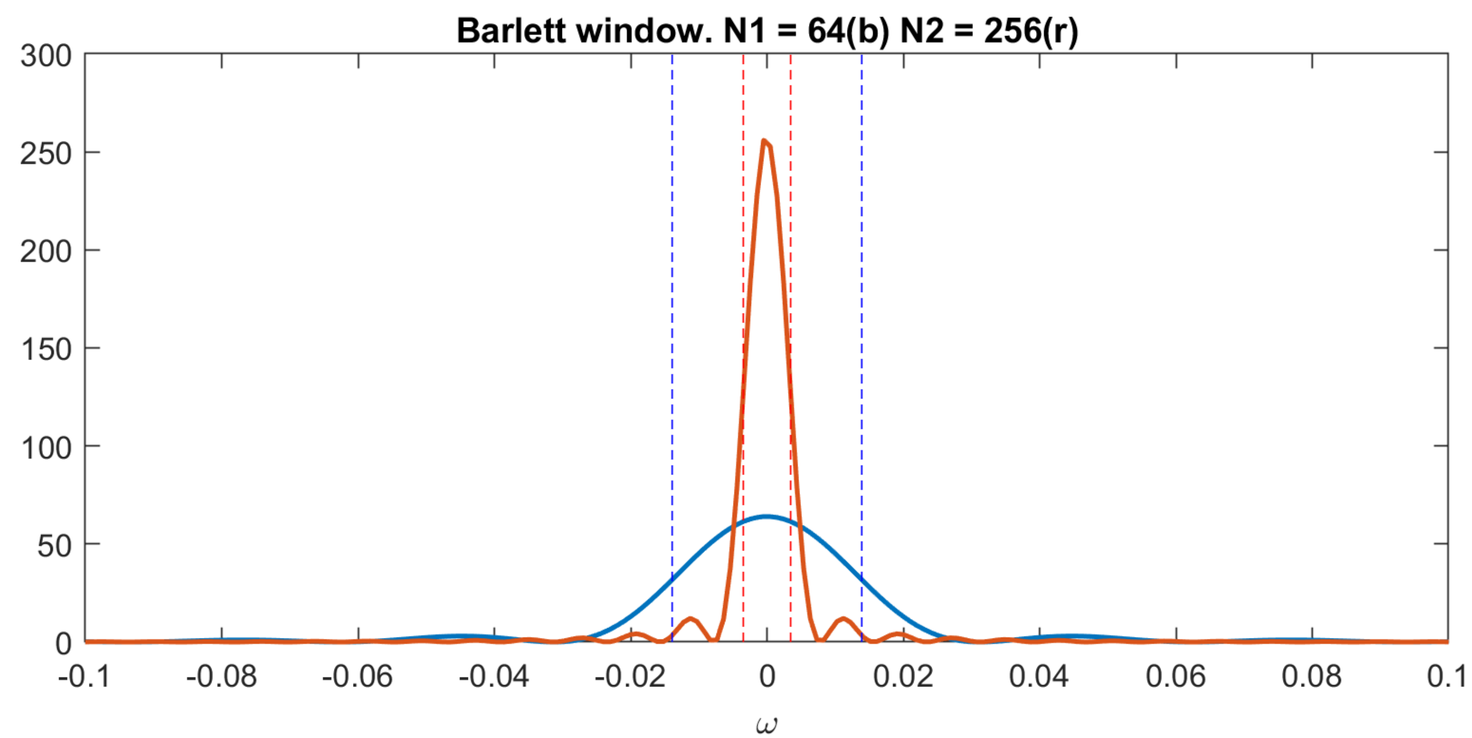
- It gives an expected periodogram:

$$E\{\hat{P}_{per}(e^{j\omega})\} = \sum_{k=-\infty}^{\infty} r_x[k] \cdot w_B[k] \cdot e^{-jk\omega}$$

- And in Fourier: $E\{\hat{P}_{per}(e^{j\omega})\} = P_x(e^{j\omega}) * W_B(e^{j\omega})$

Where Barlett Window

$$W_B(e^{j\omega}) = \frac{1}{N}\left(\frac{\sin\left(\frac{N\omega}{2}\right)}{\sin\left(\frac{\omega}{2}\right)}\right)^2$$

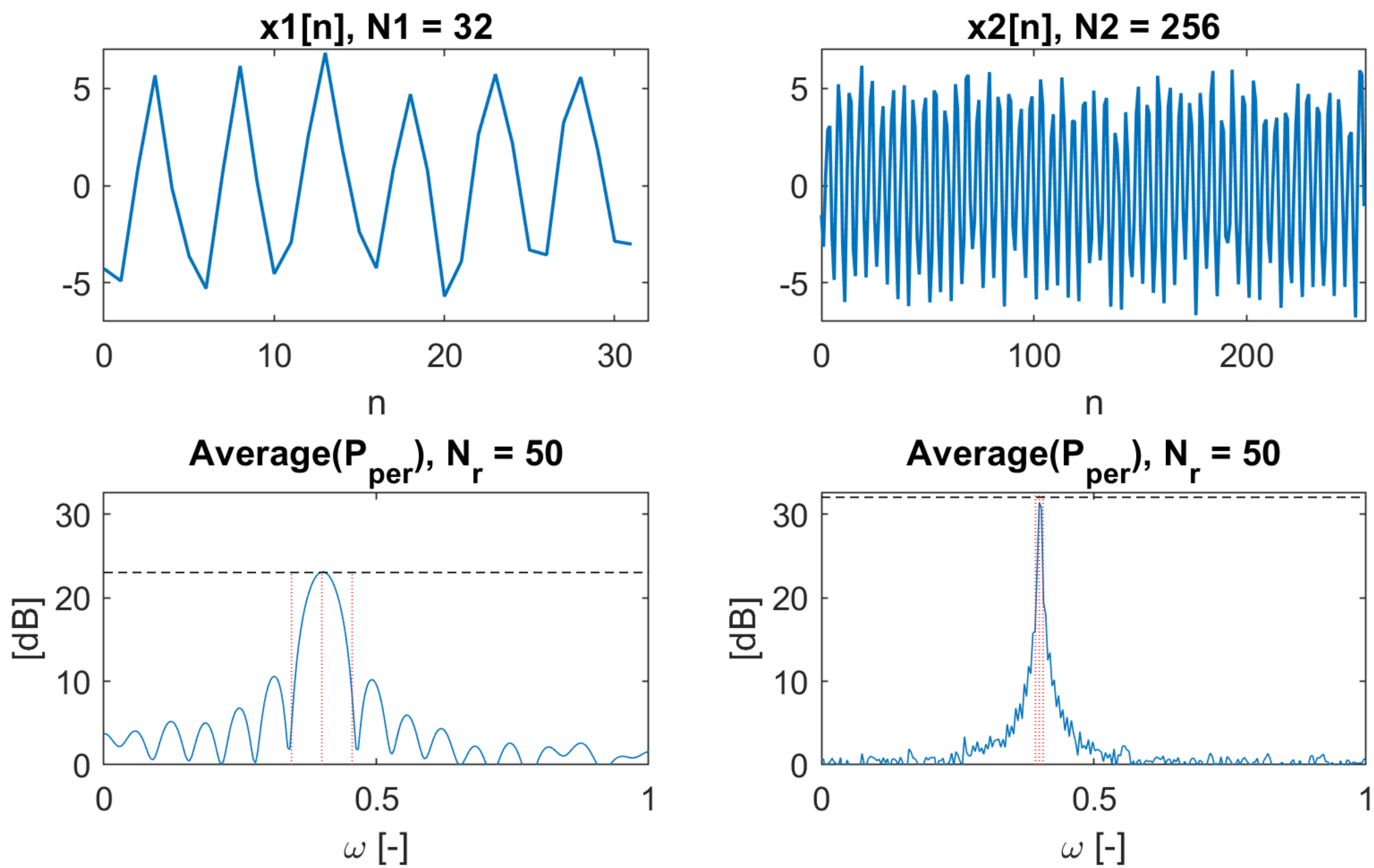


- Since the sinc-squared pulse converge to a Dirac as N goes to infinity, then the <u>periodogram is asymptotically unbiased</u>

$$\boxed{\lim_{N\to\infty} E\{\hat{P}_{per}(e^{j\omega})\} = P_x(e^{j\omega})}$$

Periodogram (Bias)

- This implies that the periodogram of a pure sine wave, will not be an impulse but the Barlett window centered at the right frequency!



⇨ *Higher is N, thinner is the main lobe and lower are the side lobes!*

Periodogram (Resolution)

- Resolution of periodogram, is ability to resolve closely spaced narrowband components which is limited by the Barlett window.
  - As the width of the main lobe increases when N decreases, there is a limit on how closely two sinusoids may be located.
  - Defined as the bandwidth of the window as its half power points (-6dB)

Periodogram → $\boxed{Res\{\hat{P}_{per}(e^{j\omega})\} = 0.89\frac{2\pi}{N}}$ $\omega \in [-\pi, \pi]$

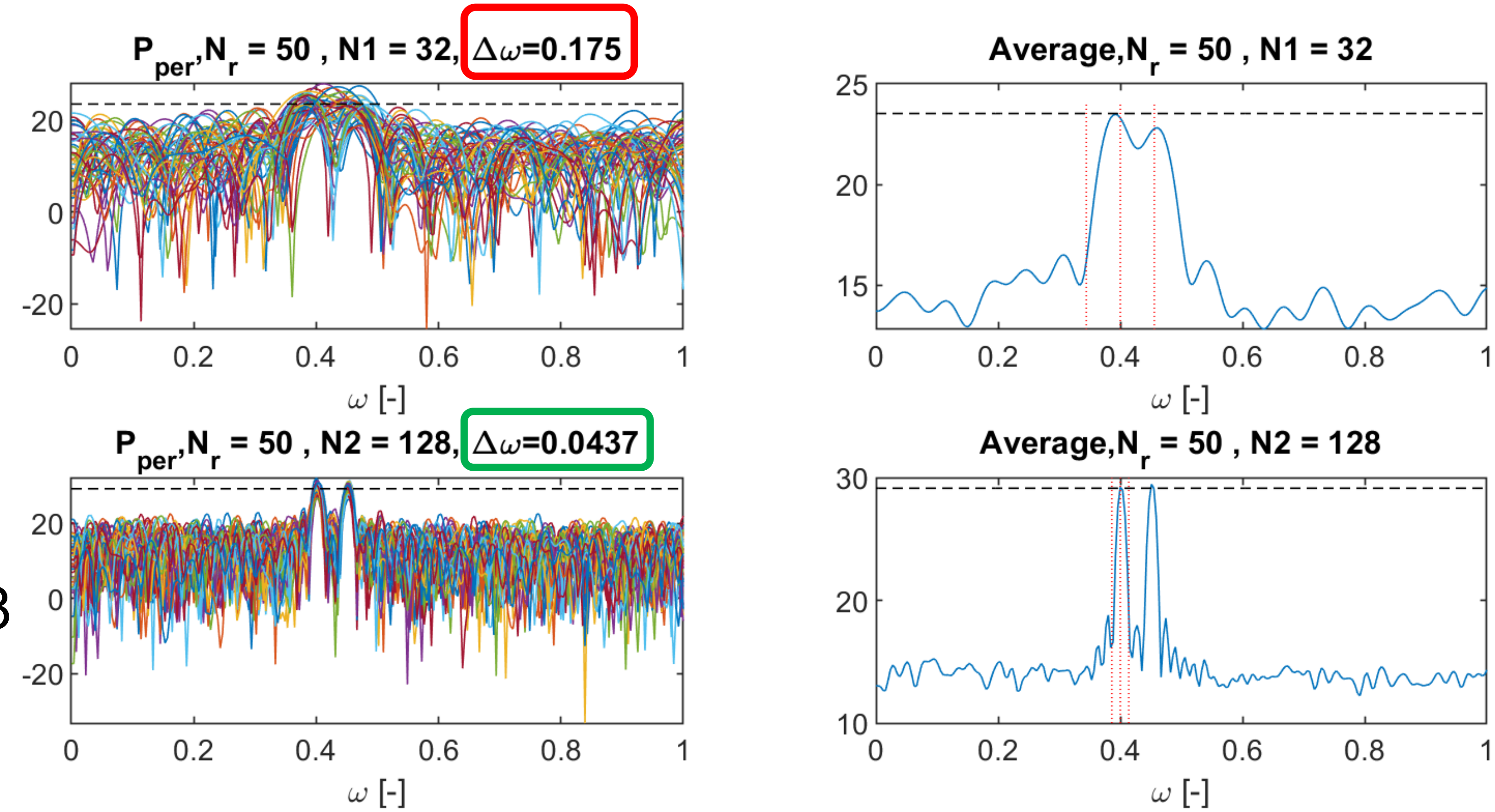
Resolution too high!

N=32

2 sines waves w=[0.4π,0.45π] with random noise. Need resolution < 0.05 π ≈ 0.16

N=128

Resolution ok!

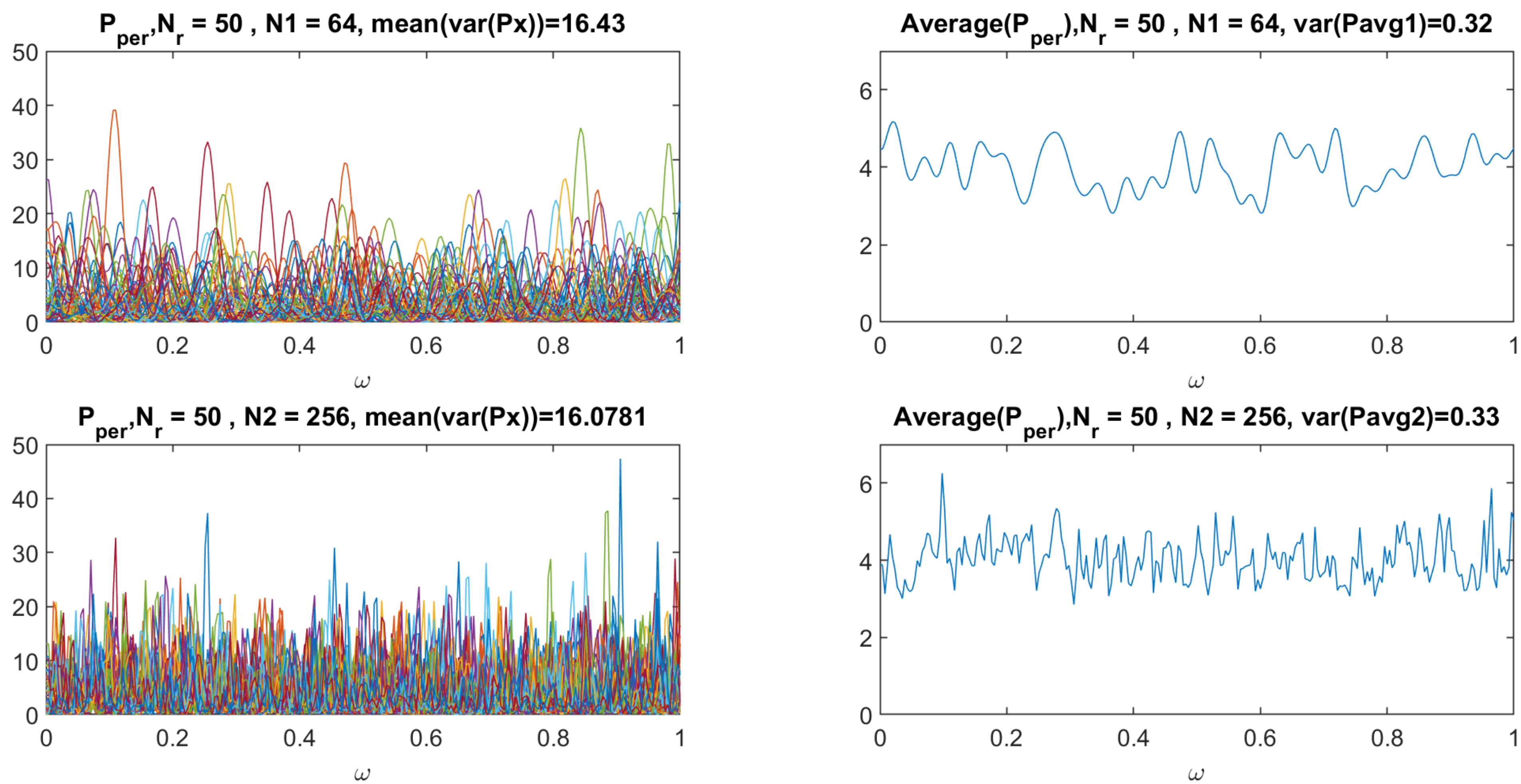


Periodogram (variance)

- As seen, the variance of the periodogram does not go to 0 when $N \to \infty$.

$$Var\{\hat{P}_{per}(e^{j\omega})\} \approx P_x^2(e^{j\omega})$$

- So, periodogram is **not a consistent estimate** of the power spectrum
- Example: White Gaussian noise $\mu = 0, \sigma = 2 \Rightarrow P_x = \sigma^2 = 4$



Modified Periodogram

- Instead of using a rectangular window $w_R[n]$ to $x[n]$ to make it finite length, we can use other kind of windows.
- This change of windows gives us the possibility to smooth the periodogram:
  - Rectangular window has a narrow main lobe compare to others windows
  - Rectangular has large side lobes that may mask weak narrowband components
- Periodogram:

$$\hat{P}_{per}(e^{j\omega}) = \frac{1}{N}\left|X_N(e^{j\omega})\right|^2 = \frac{1}{N}\left|\sum_{n=-\infty}^{\infty} x[n]\overbrace{w_R[n]}^{\text{Rectangular window}}e^{-jn\omega}\right|^2$$

- Asymptotically unbiased

$$E\{\hat{P}_{per}(e^{j\omega})\} = \frac{1}{N}P_x(e^{j\omega}) * \underbrace{\left|W_R(e^{j\omega})\right|^2}_{= N \cdot W_B \text{ (Barlett)}}$$

$$W_R(e^{j\omega}) = \frac{\sin(N\omega/2)}{\sin(\omega/2)}e^{-j(N-1)\omega/2}$$

Modified Periodogram

- The modified periodogram is simply to use a general window $w[n]$.

$$\boxed{\hat{P}_M(e^{j\omega}) = \frac{1}{NU} \cdot \left| \sum_{n=-\infty}^{\infty} x[n] \cdot w[n] \cdot e^{-jn\omega} \right|^2}$$

Normalization

General window

$$U = \frac{1}{N} \sum_{n=0}^{N-1} |w[n]|^2 \quad \Rightarrow$$

For a rectangular window:

$$U = 1$$

Modified Periodogram (performance)

- Also asymptotically **unbiased**

$$E\{\hat{P}_M(e^{j\omega})\} = \frac{1}{NU} P_x(e^{j\omega}) * \left|W(e^{j\omega})\right|^2$$

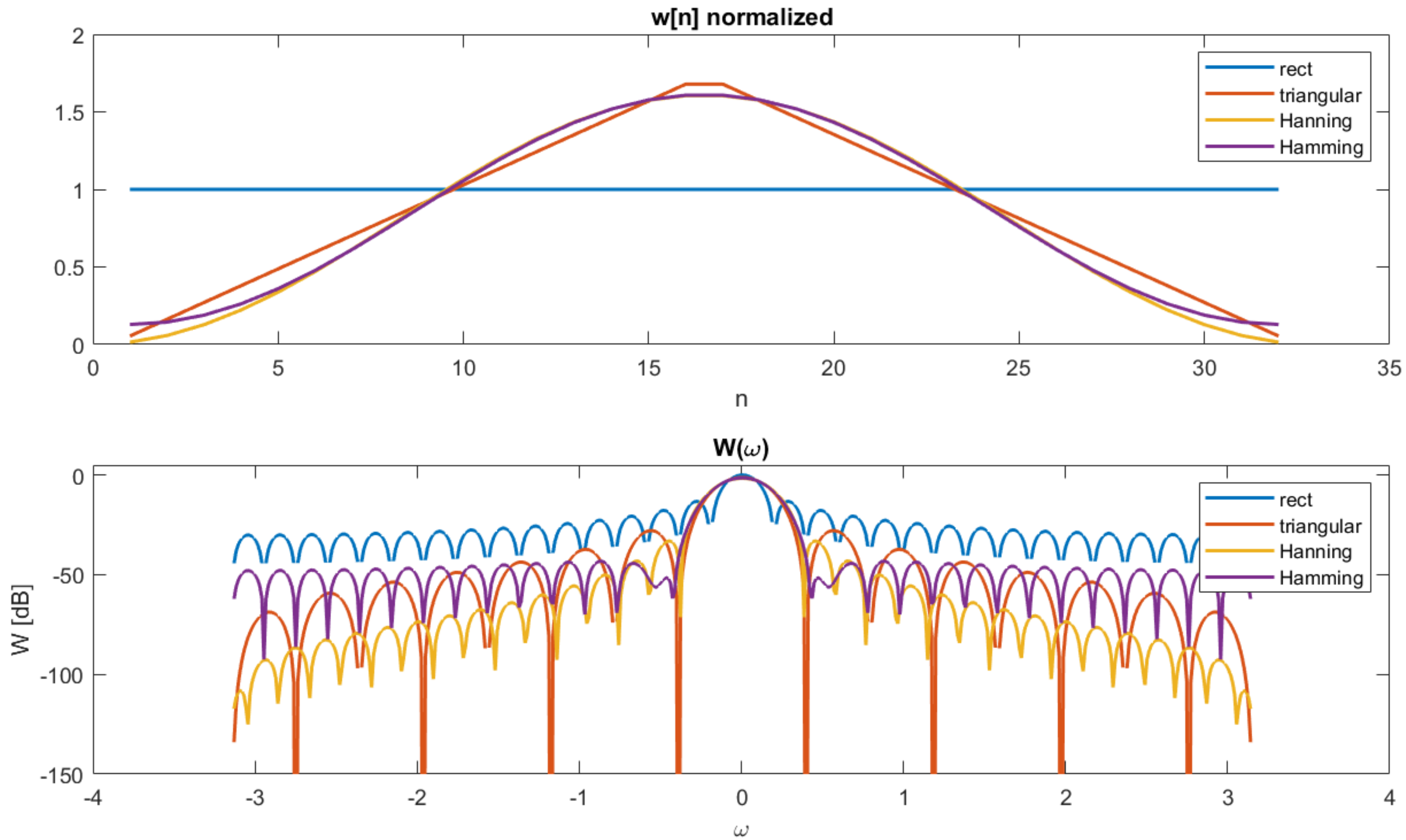
For rectangular window (periodogram) U=1 $\Rightarrow$ $\hat{P}_{per}(e^{j\omega}) = \hat{P}_M(e^{j\omega})$

- Modified periodogram is simply the periodogram of a window data sequence, the **variance** of $\hat{P}_M(e^{j\omega})$ will be approximately the same as periodogram:

$$Var\{\hat{P}_M(e^{j\omega})\} \approx \hat{P}_x^2(e^{j\omega})$$

Windows

- Commonly used windows have a wider main lobes but smaller sidelobe

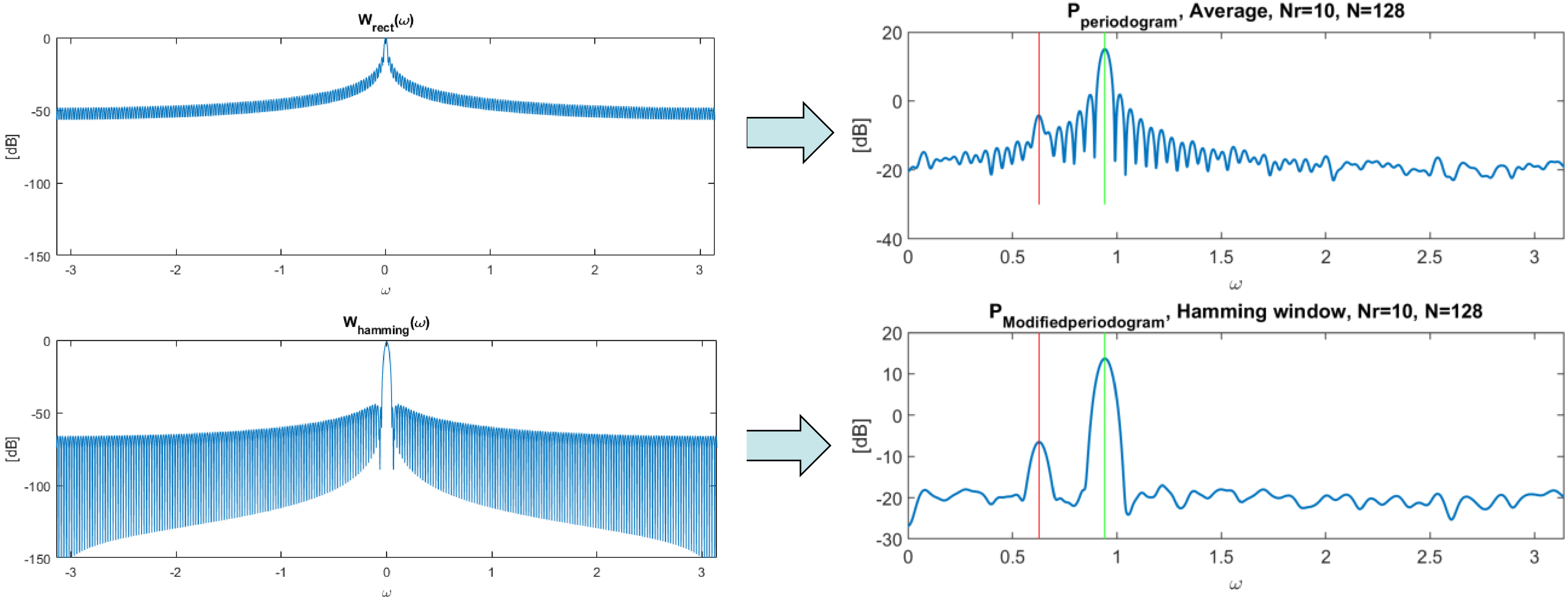


Window	Sidelobe Level (dB)	3 dB BW $(\Delta\omega)_{3dB}$
Rectangular	-13	$0.89(2\pi/N)$
Bartlett	-27	$1.28(2\pi/N)$
Hanning	-32	$1.44(2\pi/N)$
Hamming	-43	$1.30(2\pi/N)$
Blackman	-58	$1.68(2\pi/N)$

Modified Periodogram (example)

- Example of application: random process with 2 closes sinusoids (N=128)

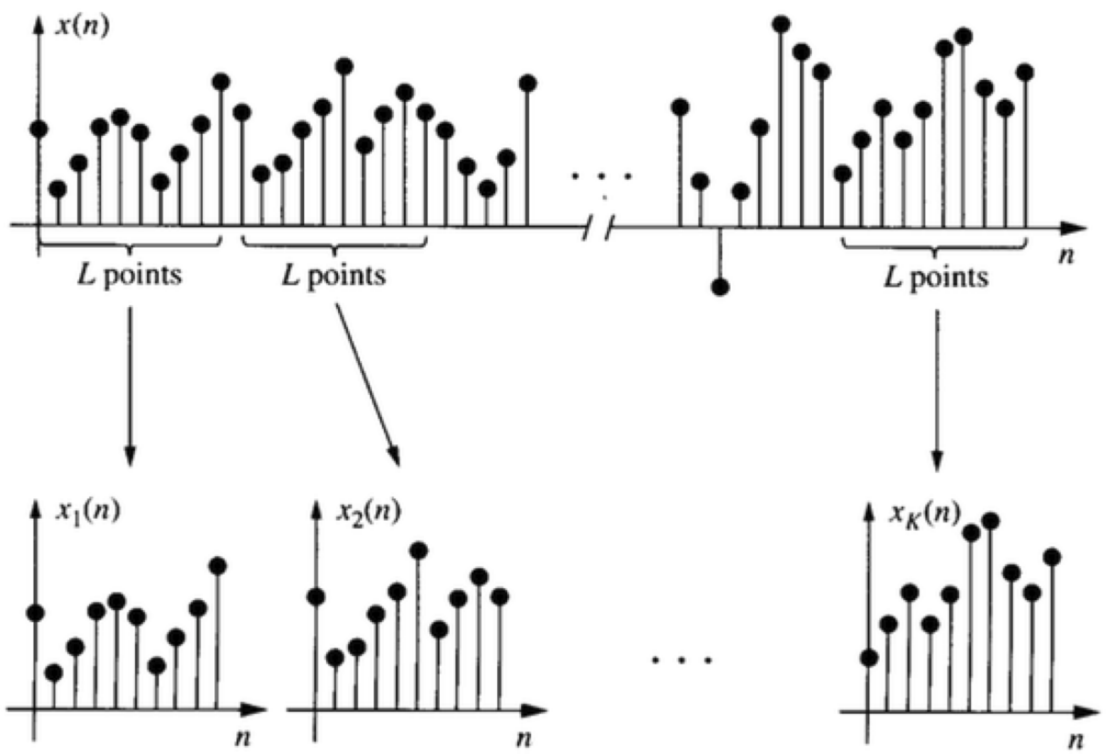
$$x[n] = 0.1 \cdot \sin(n\omega_1 + \phi_1) + \sin(n\omega_2 + \phi_2) + v[n] \qquad \omega_1 = 0.2\pi,\ \omega_2 = 0.3\pi$$



Barlett's (Averaging)

- As seen, periodogram and modified periodogram are not consistent, ie variance does not go to 0 as $N \to \infty$.
- Solution is the Barlett's method.

Segment a signal in K segments of length L. Where $\mathrm{N} = K \cdot L$



One segment:

$$\hat{P}_{per}^{(i)}(e^{j\omega}) = \frac{1}{L}\left|\sum_{n=0}^{L-1} x_i[n]\, e^{-j\omega n}\right|^2$$

Barlett (average of segment):

$$\boxed{\hat{P}_B(e^{j\omega}) = \frac{1}{K}\sum_{i=0}^{K-1} \hat{P}_{per}^{(i)}(e^{j\omega}) = \frac{1}{N}\sum_{i=0}^{K-1}\left|\sum_{n=0}^{L-1} x[n + i \cdot L]\, e^{-j\omega n}\right|^2}$$

Barlett's (performance)

- It gives also an asymptotically unbiased estimate:

$$E\{\hat{P}_B(e^{j\omega})\} = P_x(e^{j\omega}) * W_B(e^{j\omega})$$

- But it gives a consistent estimator ($K \to \infty \Rightarrow Var\{\hat{P}_B(e^{j\omega})\} = 0$)

$$Var\{\hat{P}_B(e^{j\omega})\} \approx \frac{1}{K} P_x^2(e^{j\omega})$$

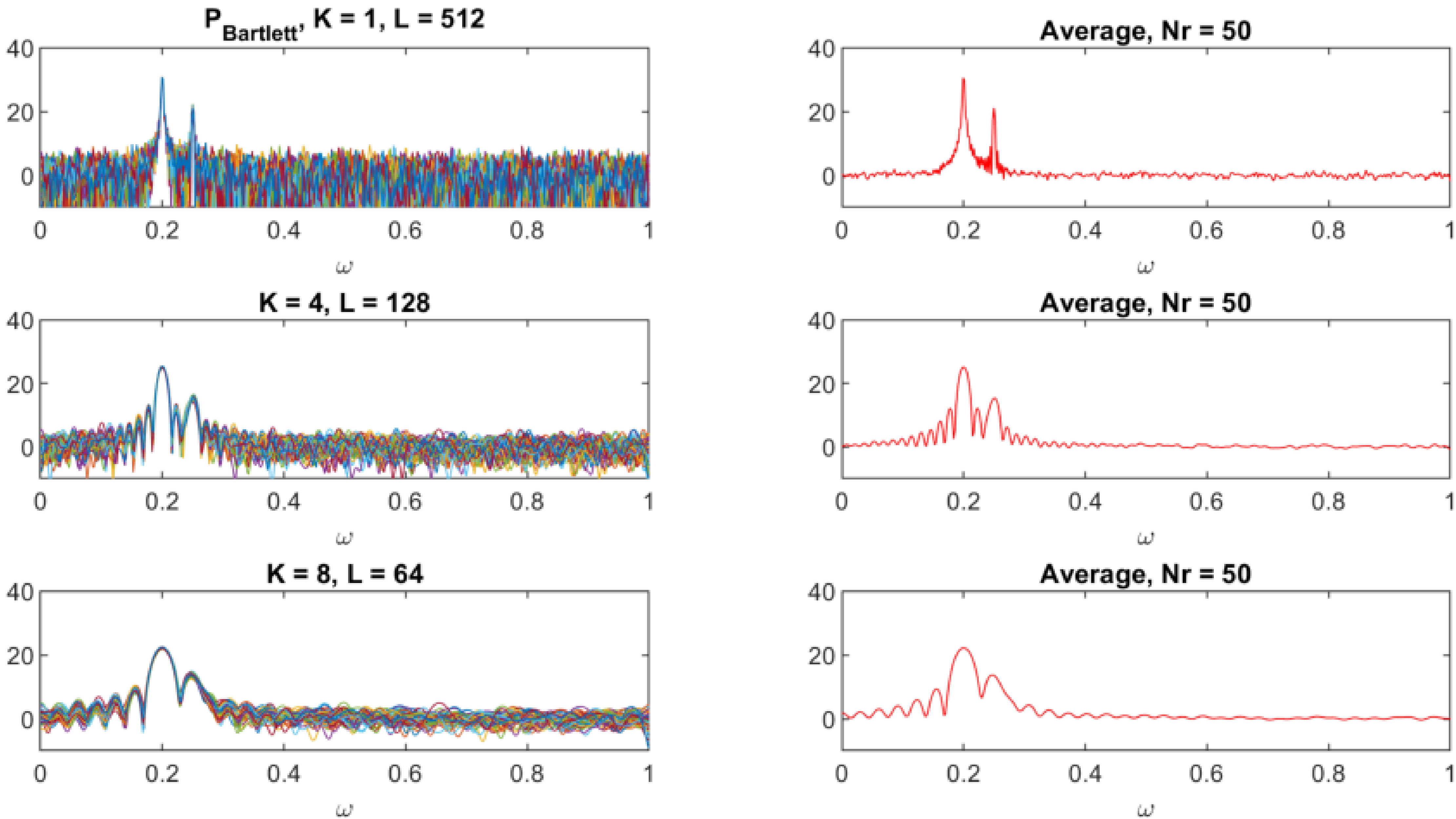
- However, the resolution increase because Barlett window is larger :

$$\Delta\omega = 0.89K\frac{2\pi}{N}$$

Barlett's (example)

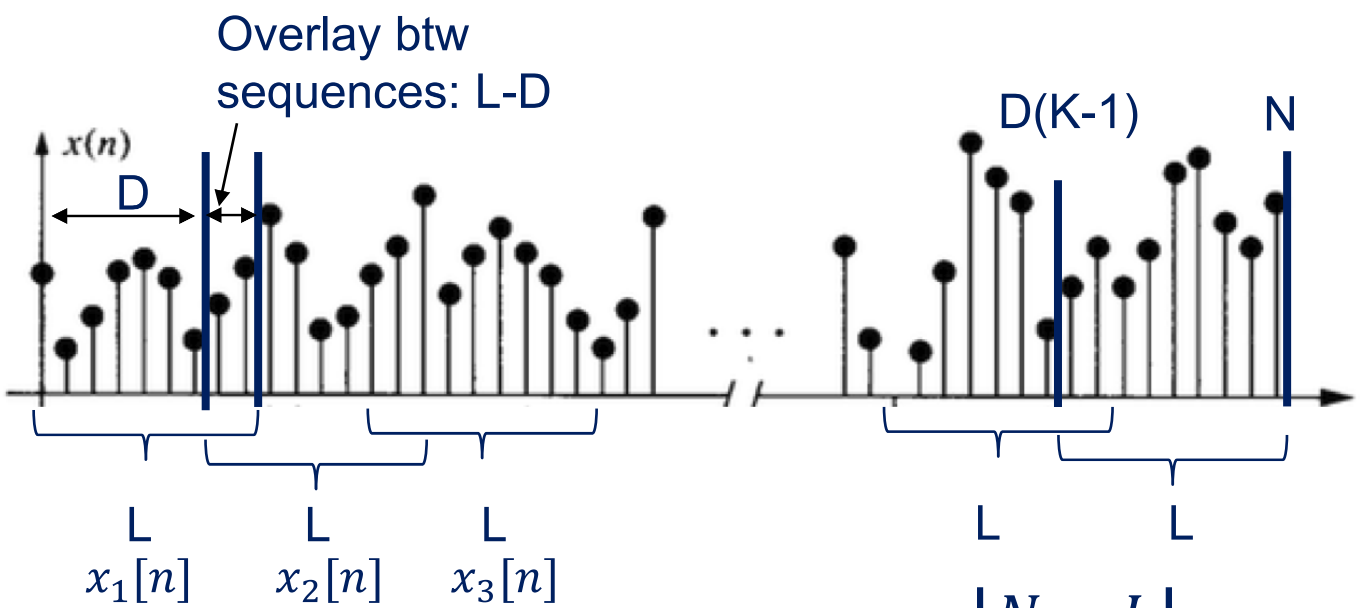
- Example : random process of two sinusoids in unit variance noise (N=512)

$$x[n] = \sqrt{10} \cdot \sin(n\omega_1 + \phi_1) + \sin(n\omega_2 + \phi_2) + v[n] \qquad \omega_1 = 0.2\pi,\ \omega_2 = 0.25\pi$$



Welch's (modified + overlap)

- In 1967, Welch proposed two modifications to Barlett's method:
  1. Allow the sequences to overlap
  2. Allow data window $w[n]$ to be applied to each sequence (as modified periodograms)



K is the number of sequence to cover all the signals: $K = \left\lfloor \frac{N-L}{D} \right\rfloor + 1$

D can be defined as a % of L (Ov : Overlay): $K = \left\lfloor \frac{N-L}{L(1-Ov)} \right\rfloor + 1$

Welch's periodogram is simply : $\boxed{\hat{P}_W(e^{j\omega}) = \frac{1}{K} \sum_{i=0}^{K-1} \hat{P}_M^{(i)}(e^{j\omega})}$

Welch's (performance)

- It gives also an asymptotically unbiased estimate:

$$E\{\hat{P}_W(e^{j\omega})\} = \frac{1}{LU} P_x(e^{j\omega}) * \left|W(e^{j\omega})\right|^2$$

- Variance is more difficult to compute since, with overlapping sequences, the modified periodograms **cannot be assumed to be uncorrelated**. With 50% overlap, the variance is approximately:

$$Var\{\hat{P}_W(e^{j\omega})\} \approx \frac{9}{8K} P_x^2(e^{j\omega})$$

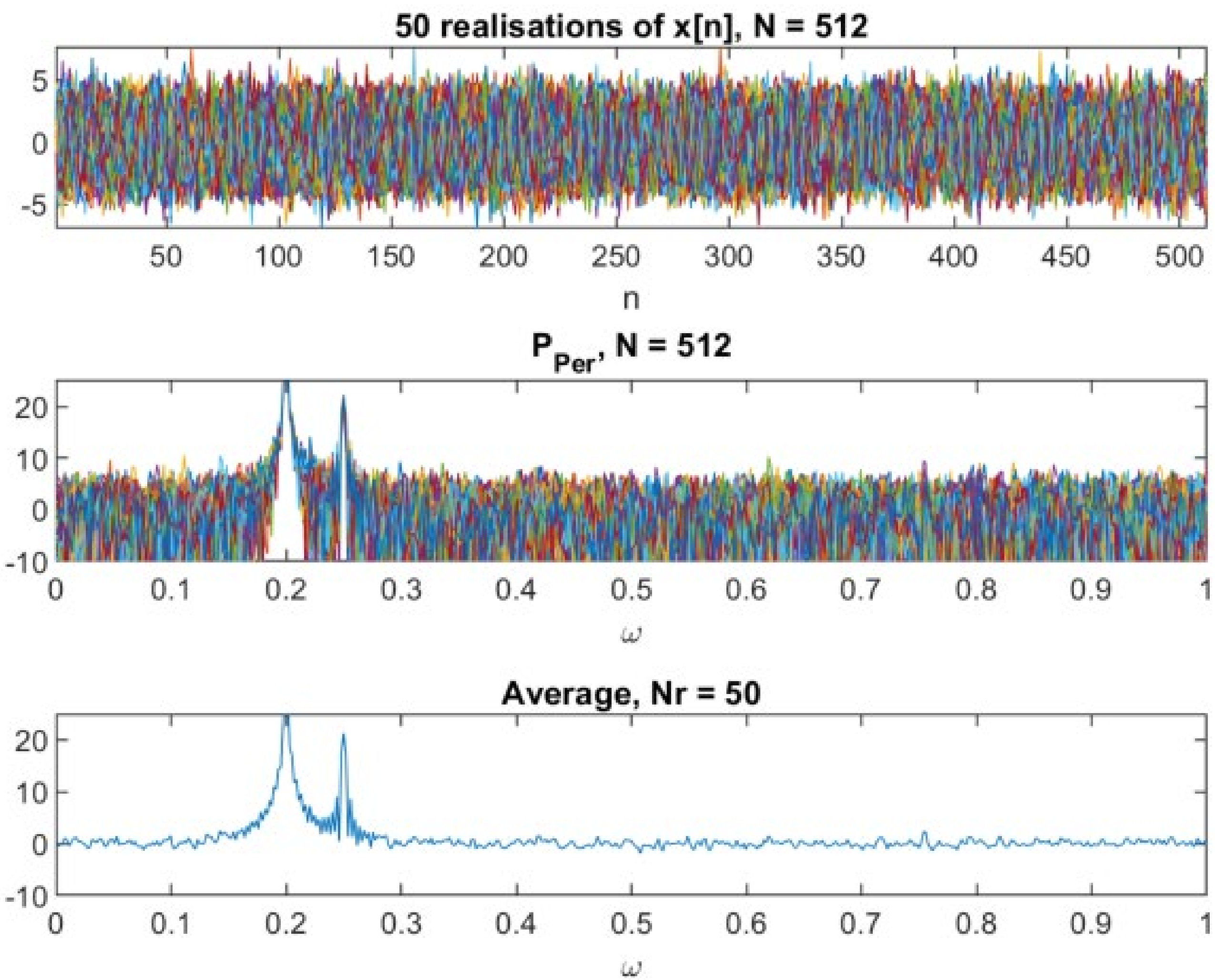
So consistent as $Var \to 0 \; if \; K \to \infty$

Welch's (example)

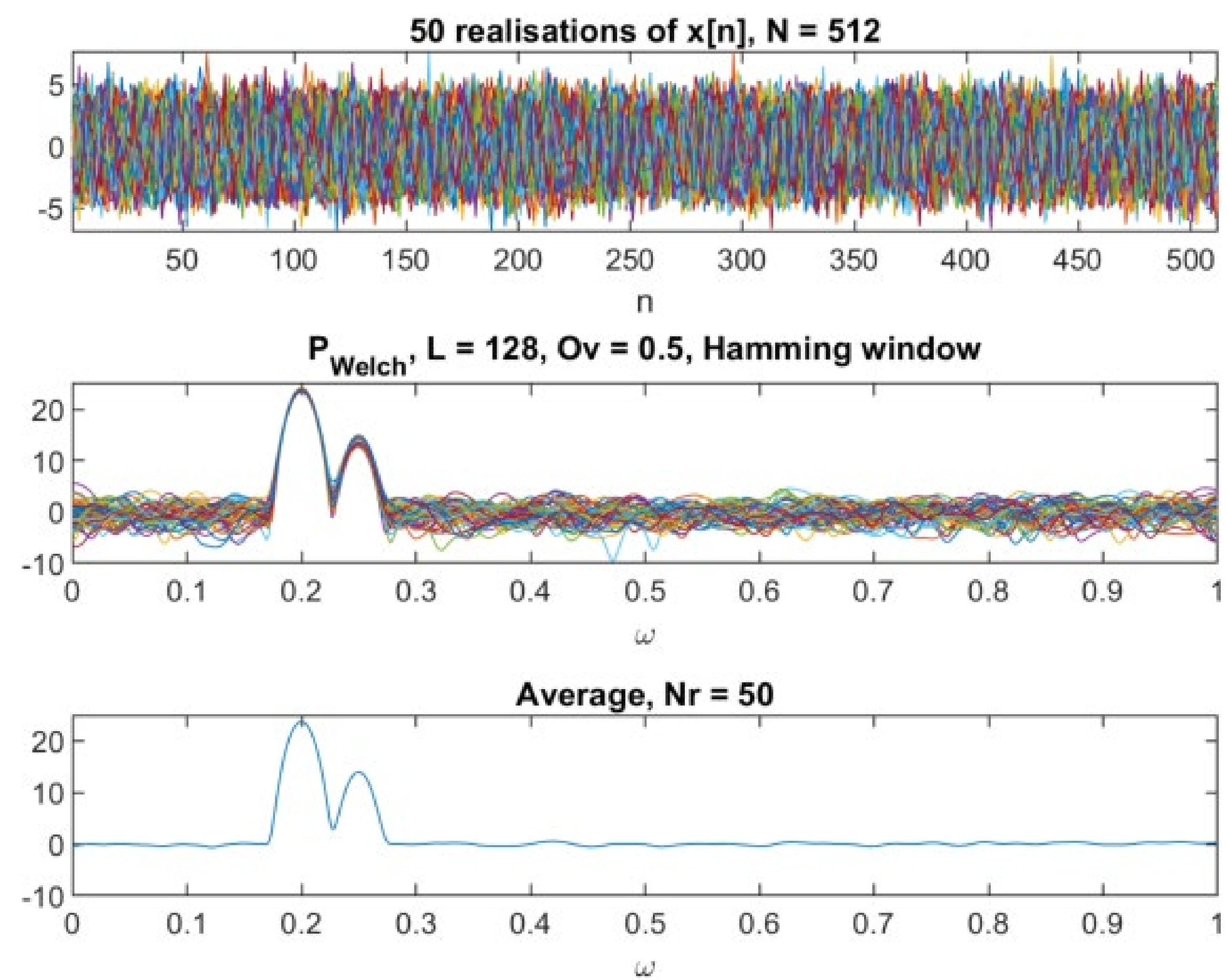
- Example : random process of two sinusoids in unit variance noise (N=512)

$$x[n] = \sqrt{10} \cdot \sin(n\omega_1 + \phi_1) + \sin(n\omega_2 + \phi_2) + v[n] \qquad \omega_1 = 0.2\pi,\ \omega_2 = 0.25\pi$$

Periodogram



Welch

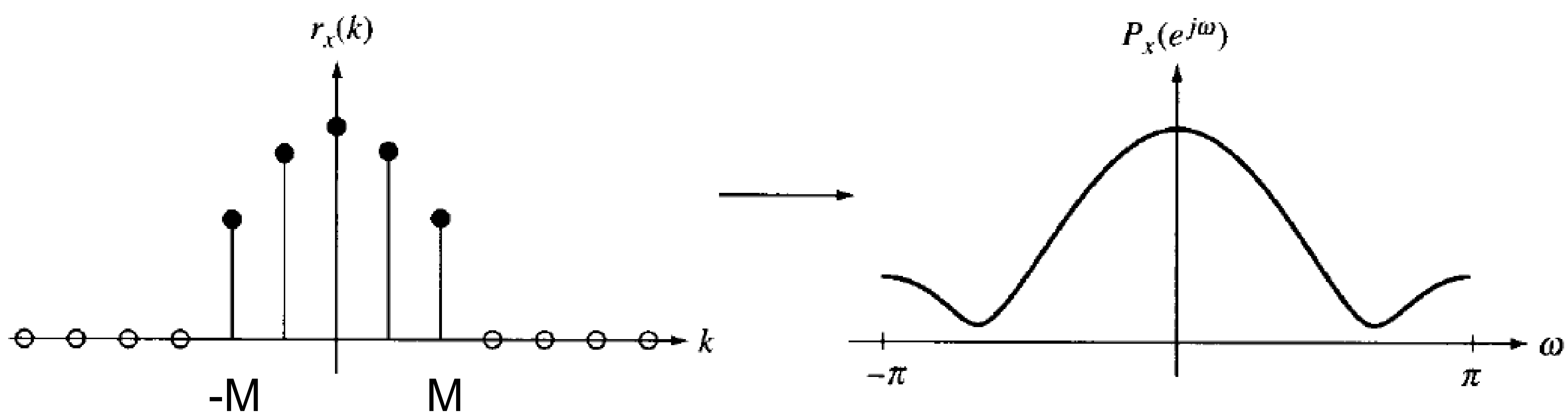


Example826.mlx

Others non parametric method

- Blackman-Tukey Method
  - Window on the autocorrelation as the extreme autocorrelation values are noisy $r_x(k)$ where $|k| \approx N$

$$\hat{P}_{BT}(e^{j\omega}) = \sum_{k=-M}^{M} \hat{r}_x(k) w(k) e^{-jk\omega}$$



Others non parametric method

- Minimum Variance Spectrum Estimation
  - Design a bank of bandpass filter with a bandwidth Δ and a center frequency ω_i and filter the signal with each filters.

$$\hat{P}_x\left(e^{j\omega_i}\right) = \frac{E\{|y_i(n)|^2\}}{\Delta/2\pi}$$

- Maximum Entropy Method (MEM)
  - Extrapolation of the autocorrelation